PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila















PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

FUELING THE FUTURE OF ANIMAL NUTRITION WITH IMMUNOPEPTIDES:

Enhancing Gut Health Immunity and Sustainable Performance.

Concepcion D. Villacorta, DVM
Asst. Professor II, Institute of Veterinary Medicine
Bulacan Agricultural State College
Farm Owner/Entrepreneur

OBJECTIVES

1. Understand the link between gut health, immunity, resilience, and performance.
2. Describe how immunopeptides may support animal health and productivity.
3. Appreciate their potential role in sustainable nutrition and responsible antimicrobial stewardship.

Why Functional Nutrition Matters Now

- **DEMAND**
- **EFFICIENCY**
- **HEALTH**
- **STEWARDSHIP**
- **SUSTAINABILITY**

The nutrition question is shifting from

"How much can we feed?"
to "How can nutrition help animals remain healthy, efficient, and resilient?"

The New Nutrition Lens: Gut-Microbiota-Immune Function



Dietary inputs



Barrier + microbiota



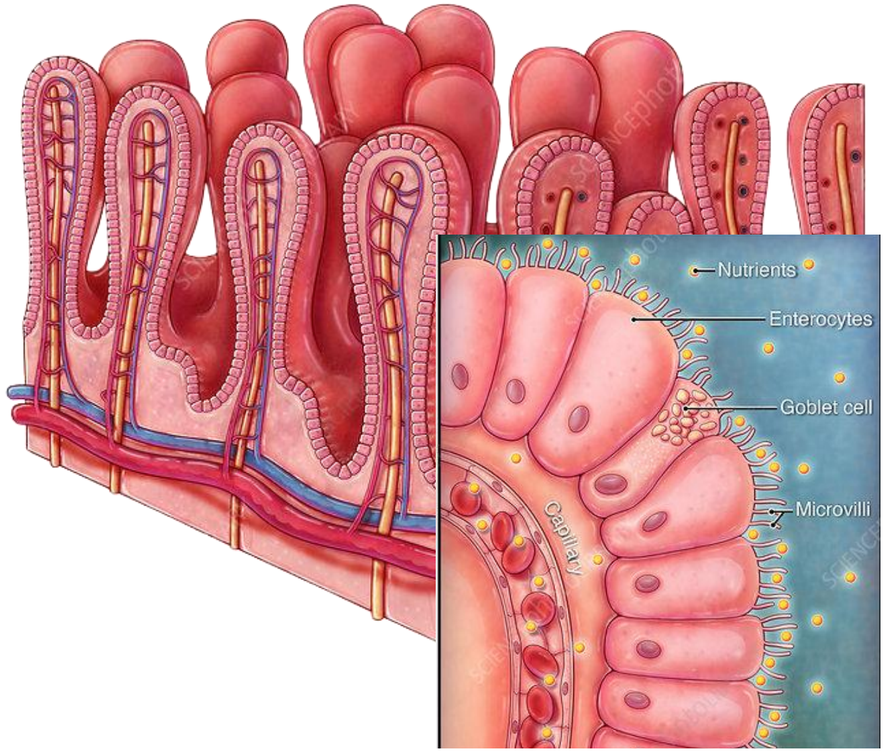
Host response



Growth + efficiency = resilience

Functional nutrition can act upstream—supporting the biological systems that make performance possible.

GUT HEALTH: The Foundation of Performance



- Barrier integrity
- Balanced microbiota
- Effective digestion & absorption
- Lower intestinal stress

Healthy gut → Nutrients can be converted into performance more efficiently.

Immunity + Resilience

Heat Stress



Weaning



Handling



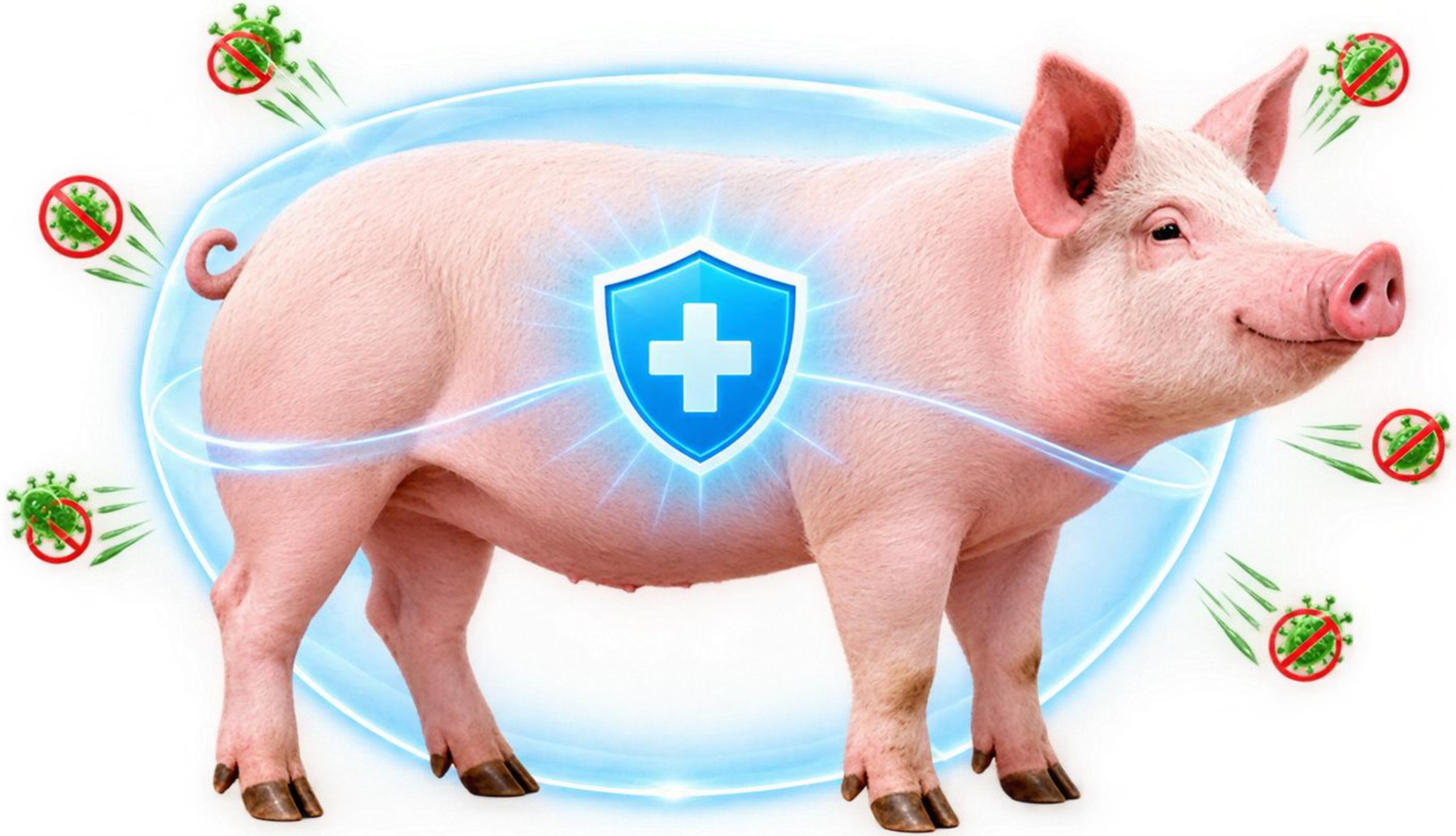
Disease Pressure

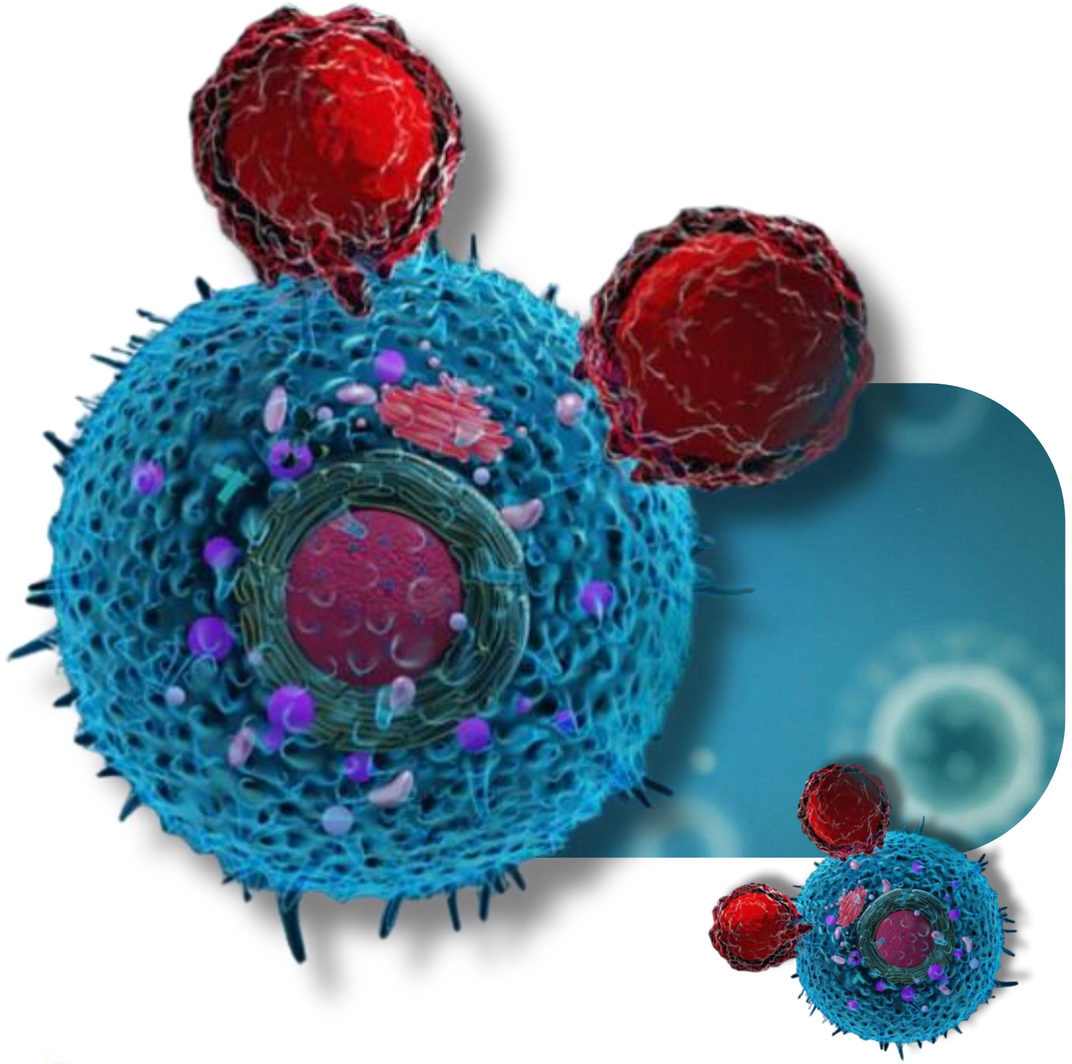


Resilience

- Appropriate Immune Response
- Inflammatory Balance
- Recovery + consistency

Healthy gut + strong immunity + resilience → BETTER PERFORMANCE

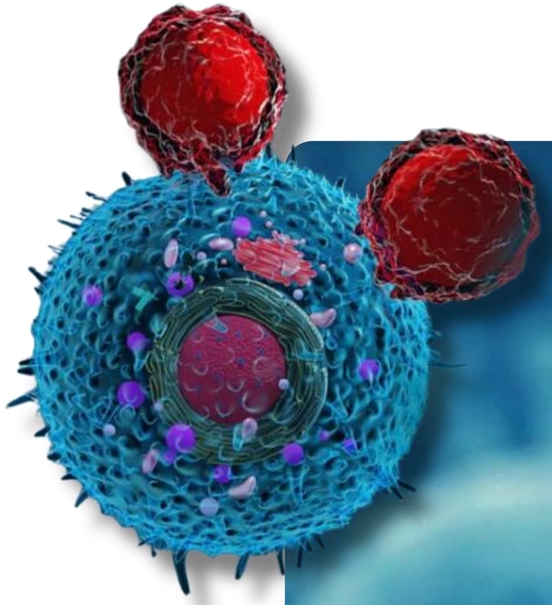




Immunopeptides

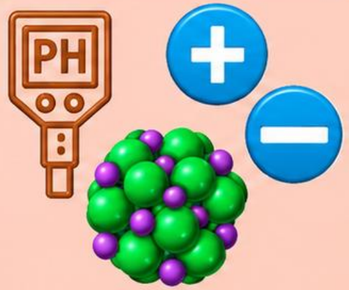
Immunopeptides are bioactive peptides capable of influencing biological processes beyond their basic nutritional value—particularly pathways related to intestinal health, microbial balance, and immune regulation.

Immunopeptides



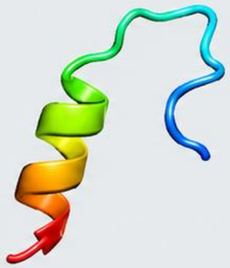


Different sources



Biochemical influence

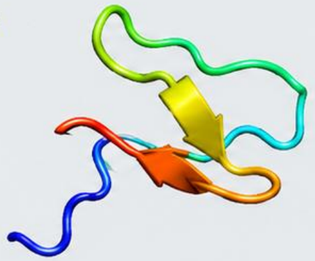
Structures of AMPs



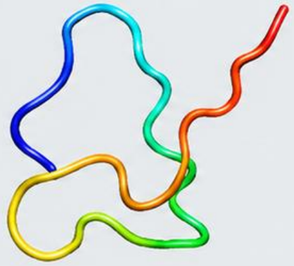
Buforin II



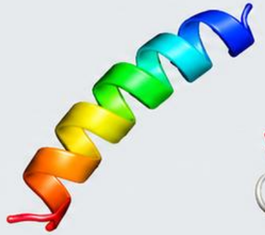
C20-Jp-Hp



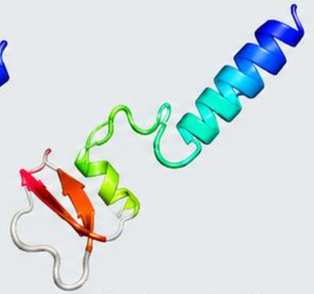
Kalata B1



α-defensin HD6



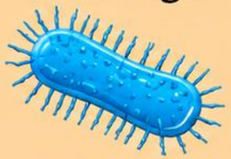
MSI-78



β-defensin 1



Antifungal



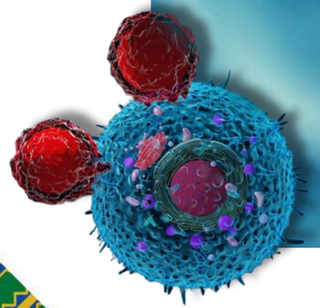
Antibacteria



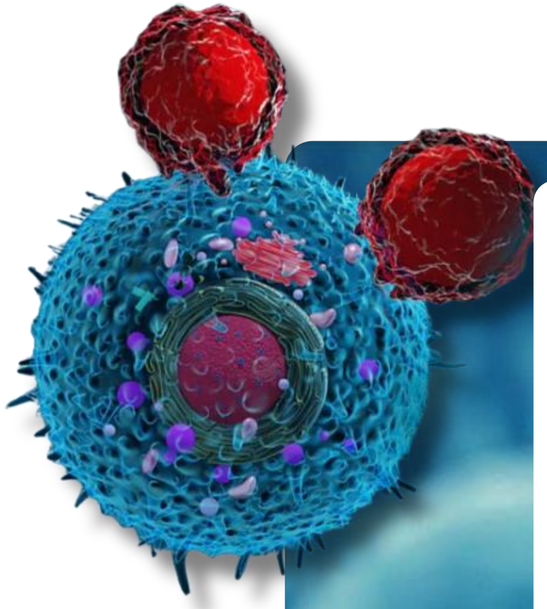
Antiparasitic

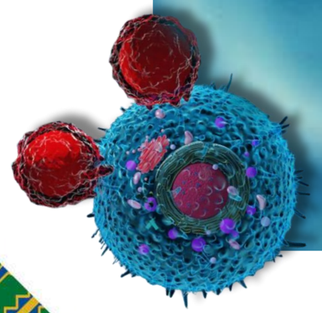


Antiviral



Immunopeptides





Antibiotics

Antibiotic resistance





Natural antimicrobial properties

Antimicrobial peptides

Low level induced resistance









1. Immune modulation
2. Improved growth performance
3. Promoted nutrient digestibility and improved gut morphology
4. Modulation of gut microbiota

Immunopeptides

SURFACTIN

- a natural immunopeptide produced by *Bacillus spp.*

CITROCIN

- a natural immunopeptide produced by *Citrobacter spp.* and *Saccharomyces spp.*

NOTE:

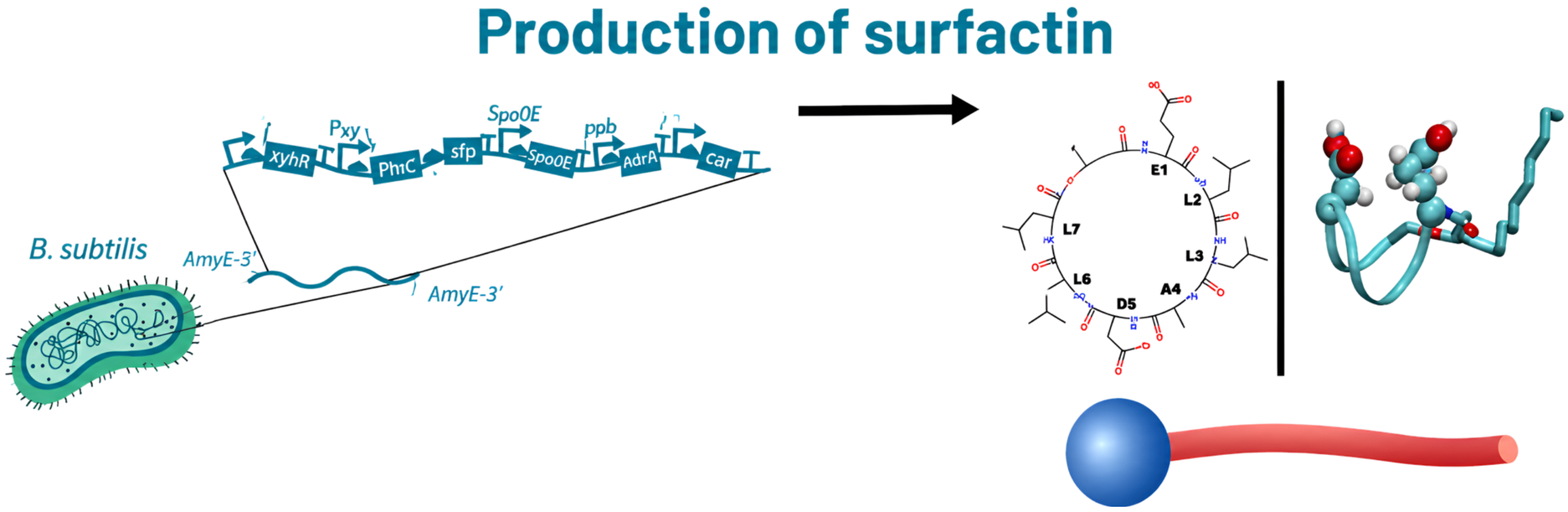
Surfactin and **Citrocin** are distinct bioactive peptides/immunopeptides that offer unique therapeutic advantages in biochemistry and immunology.

Reference: Research Gate

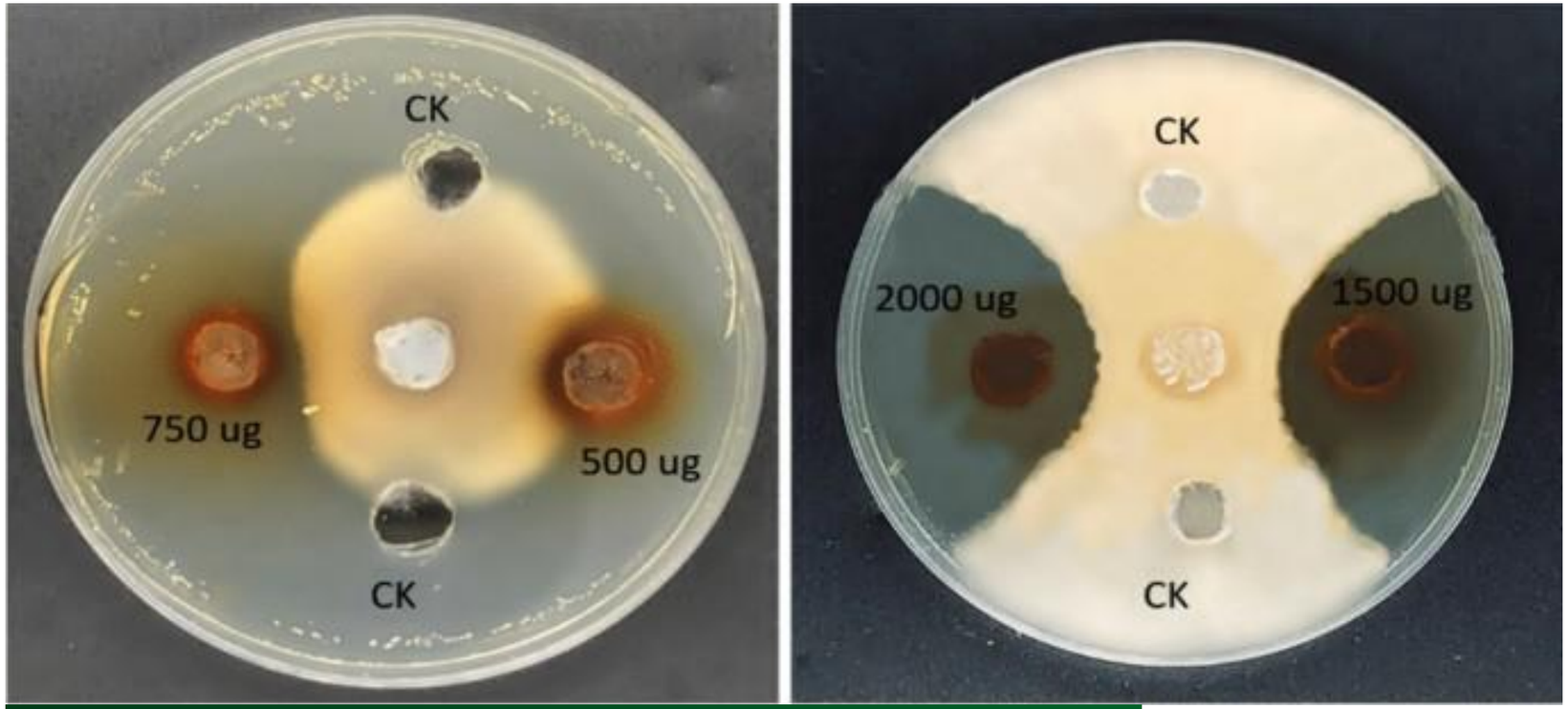
Immunopeptides

SURFACTIN

- a natural immunopeptide produced by *Bacillus spp.*



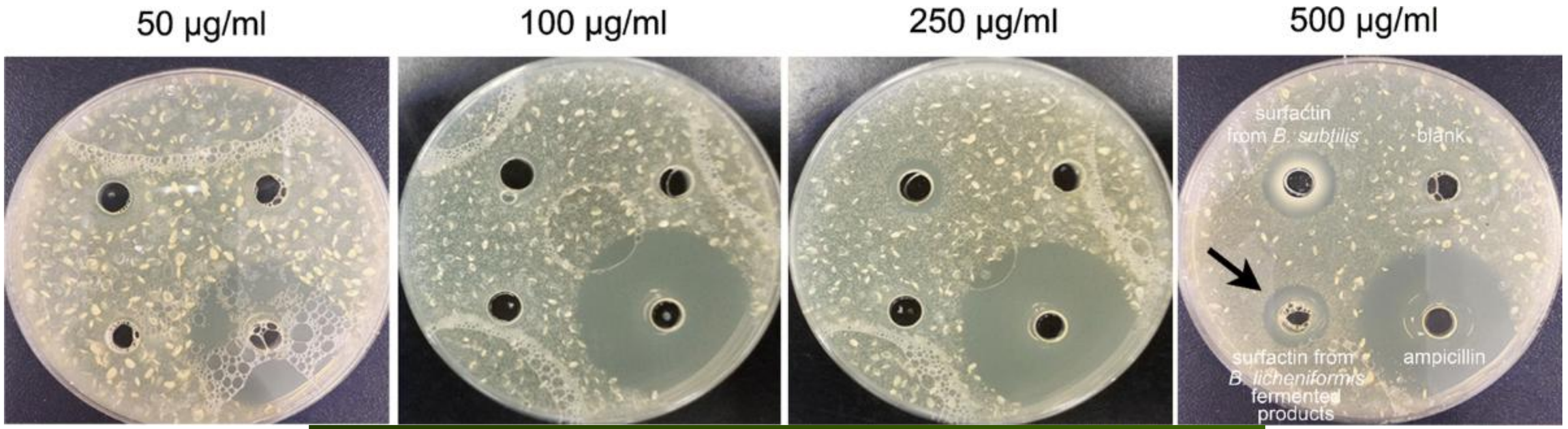
Surfactin as an antifungal alternative.



Identification of antifungal lipopeptides from *Bacillus subtilis* Sh-17 targeting *Fusarium oxysporum* f. sp. *lycopersici*

Research | Open access | Published: 03 March 2025
Volume 12, article number 29 (2025) Cite this article

Surfactin as an antibacterial alternative.



Antibacterial activity of *Bacillus* species-derived surfactin on *Brachyspira hyodysenteriae* and *Clostridium perfringens*

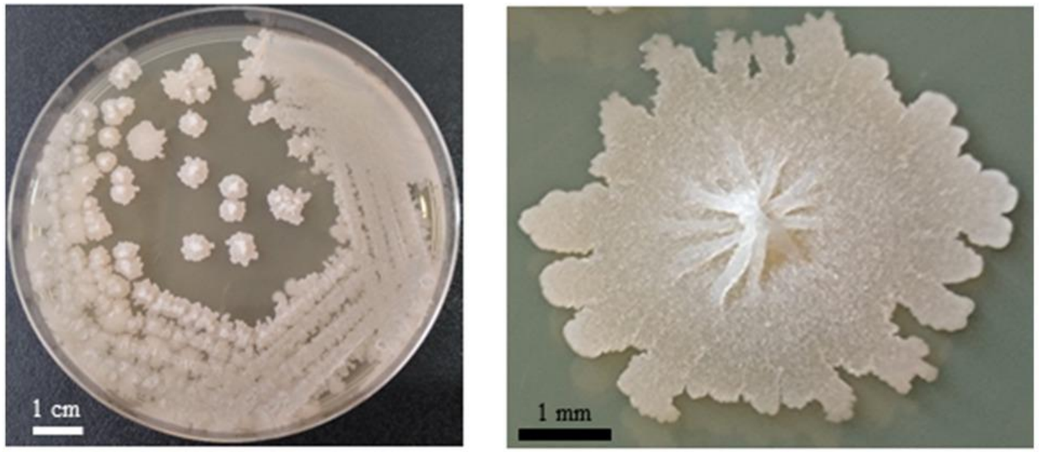
Original article | Open access | Published: 21 November 2019

Volume 9, article number 188 (2019) Cite this article

Surfactin GIT wound healing properties.

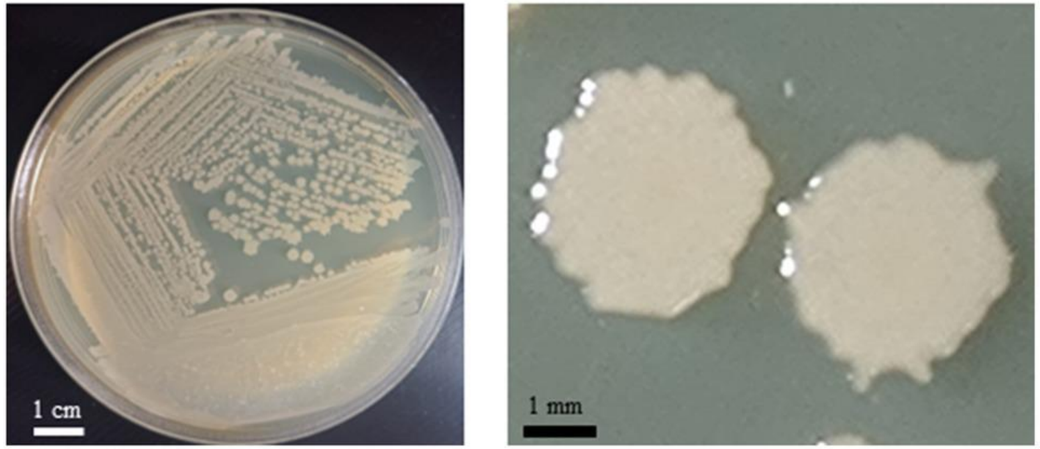
A

B. subtilis DG101



B

B. subtilis 168

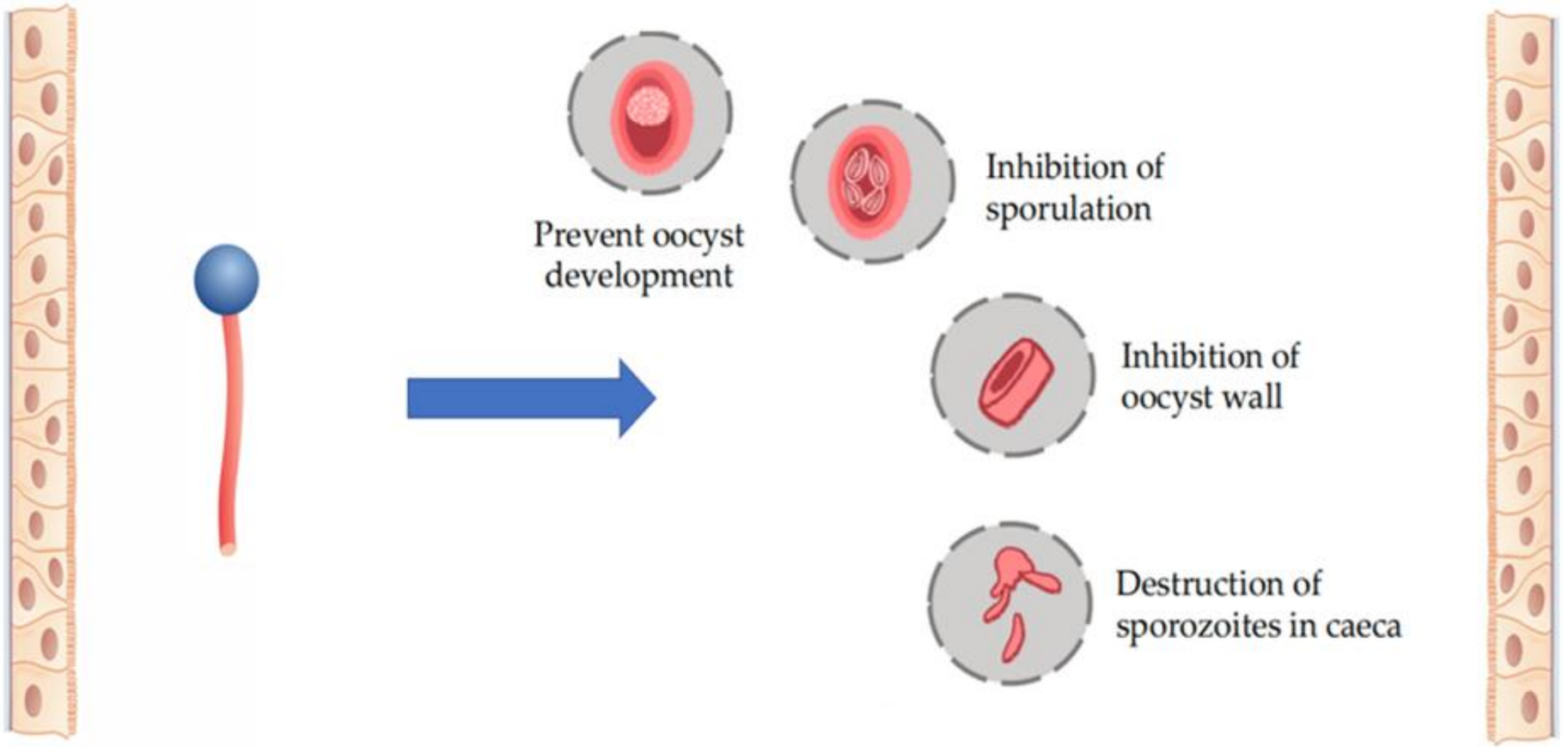


Front. Microbiol., 28 September 2023
Sec. Food Microbiology
Volume 14 - 2023 | https://doi.org/10.3389/fmicb.2023.1253480

Probiotic properties of *Bacillus subtilis* DG101 isolated from the traditional Japanese fermented food nattō

Surfactin

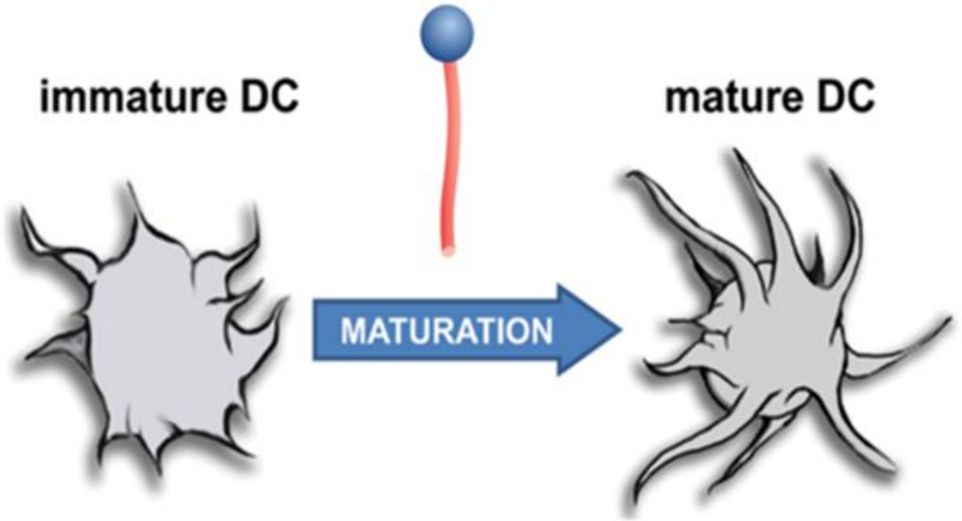
Anti-coccidial property.



Reference: https://www.thepoultrysite.com/

Surfactin Stronger immune response.

The maturation of dendritic cells (DCs) is essential for inducing downstream immune response.

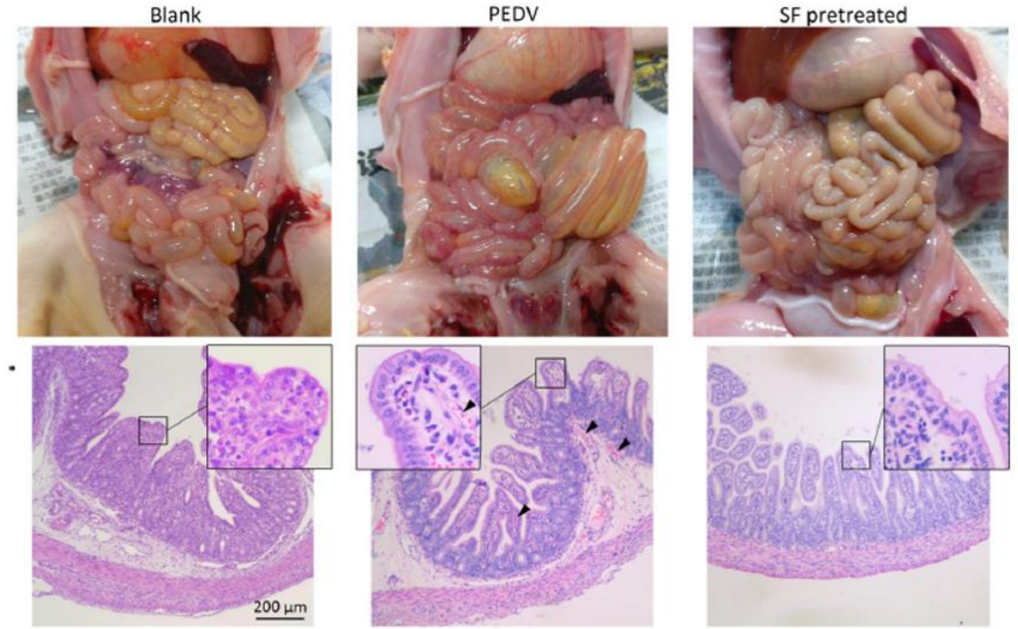


Surfactin could induce immature DCs turn to a mature state, having an enhanced migration ability and proliferation promoting ability.

Reference: Bioscience Reports (published by Portland Press / Biochemical Society)

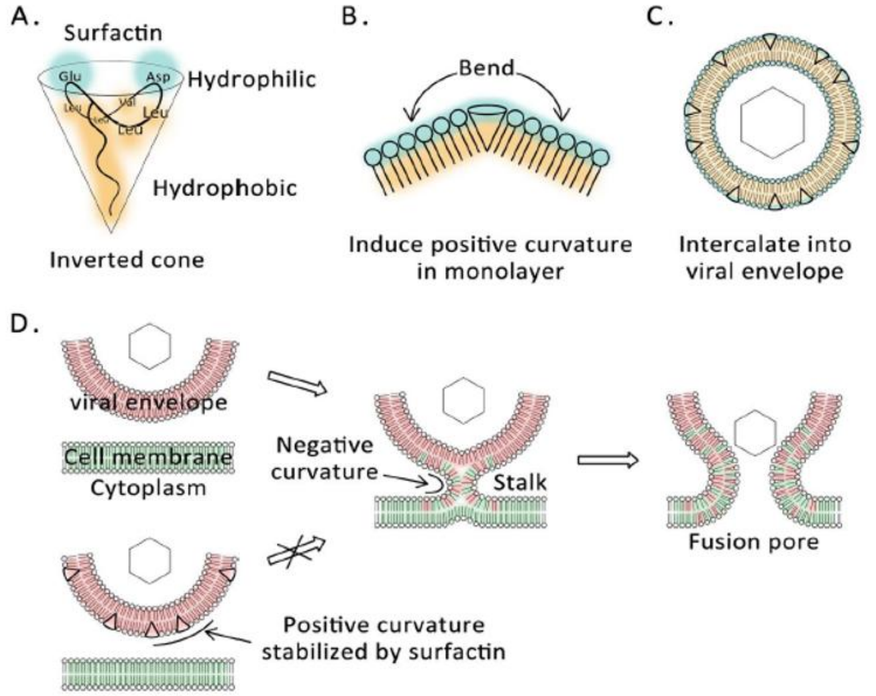
Surfactin Antiviral properties.

Oral administration of surfactin to piglets protects against PEDV infection



Anatomical photos of the abdominal cavities of the piglets.
Histopathology of the piglet jejuna. Arrowheads indicate congestion.

Surfactin inhibits membrane fusion during invasion of epithelial cells by enveloped viruses



Terms:
PEDV: Porcine Epidemic Diarrhea Virus
SF: Surfactin

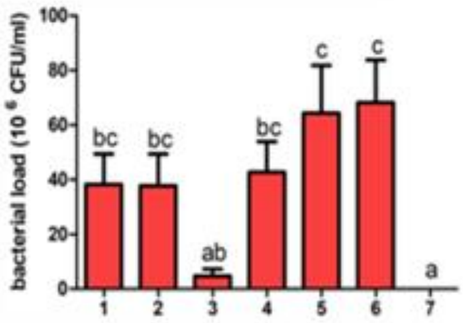
Reference: Journal of Virology (published by the American Society for Microbiology)

Surfactin In combination with Essential oils/Antibiotics.

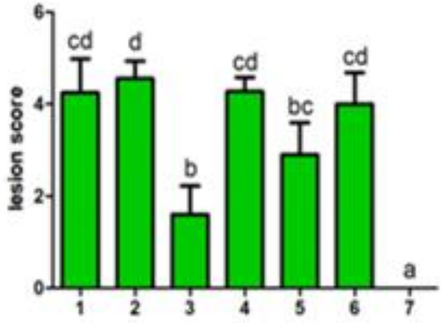
Sensitization of avian pathogenic *Escherichia coli* to amoxicillin *in vitro* and *in vivo* in the presence of surfactin

Jiaxu Liu[1], Xu Wang[1], Weiqi Shi[1], Zhuoyu Qian[2], Yongqiang Wang[1]*

Group	Treatment	Daily dose/g bodyweight, 3 times per day
1	Amoxicillin	0.02 mg/g[1]
2	Amoxicillin	0.01 mg/g[1]
3	Amoxicillin and surfactin	0.01 mg/g amoxicillin[1] + 0.01 mg/g surfactin
4	Amoxicillin and surfactin	0.005 mg/g amoxicillin[1] + 0.005 mg/surfactin
5	Surfactin	0.01 mg/g
6	Positive control	none
7	No challenged	none



Bacterial load of dead chicks and euthanized chicks' liver



Score of pathology lesion following dead chicks and euthanized chicks after

Antimicrobial effects of essential oils in combination with surfactin

Jong-Hwan Lim, Byung-Kwon Park, Myoung-Seok Kim, Mi-Hyun Hwang[1], Seung-Chun Park[1], Jae-Jin Han[2], Jong-Woo Kim[2] and Hyo-In Yun
College of Veterinary Medicine, Chungnam National University
[1]College of Veterinary Medicine, Kyungbuk National University
[2]B&C Biopharm, Yongin-si, Gyunggido





- Thyme alone active against *Escherichia coli* and *Salmonella cholerasuis*.
- Surfactin by itself showed no inhibition against both of the gram-negative bacteria.
- The combination of surfactin with the essential oils result in the enhancement of its antimicrobial activities against *Salmonella choleraesuis* and *Escherichia coli*.



Potential of surfactin to improve performance and gut health of broilers under necrotic enteritis challenge

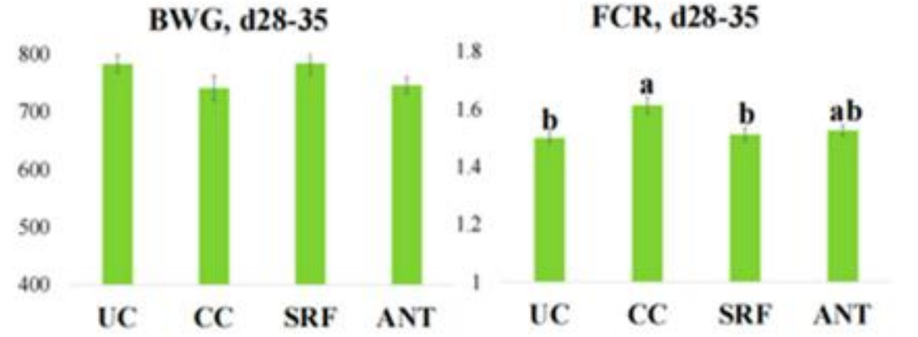
A. Kumar[1], M. Khairunnesa[1], K. Gharib-Naseri[1], P. Chen[2], L. Li[2], S.-B. Wu[1]*

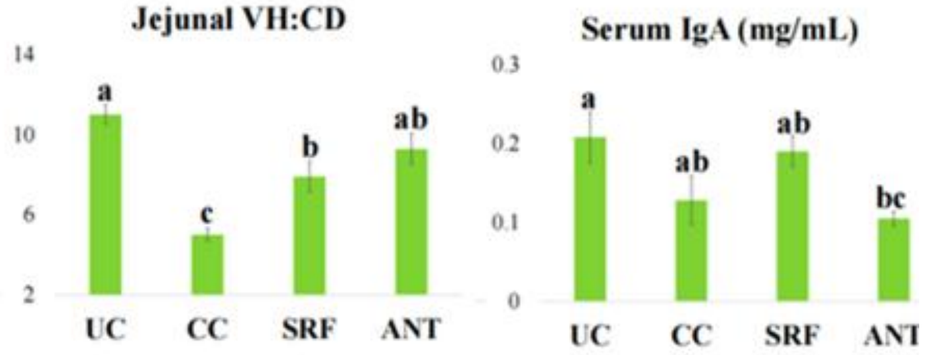
[1]School of Environmental and Rural Science, University of New England, Armidale 2351, Australia;
[2]Beijing Enhalor International Tech Co. Ltd., China

- 512 Cobb 500 chicks, 4 treatments with 8 replicates of 16 birds
- Wheat-sorghum-soybean meal based diet supplemented with phytase
- Female percentage was used as a covariate for performance data
- Birds challenged with *Eimeria* d 9 and *C. perfringens* on d14 & 15

Treatments[1]	Additives	Inclusion level, % in all phases	NE Challenge
1. UC	-	-	Unchallenged
2. CC	-	-	Challenged
3. SRF	Surfactin	0.01	Challenged
5. ANT	Antibiotics	0.027+0.05	Challenged

[1]UC: unchallenged control, CC: challenged control, SRF: CC+Surfactin at 100 g/t, and ANT: CC+Antibiotics (zinc bacitracin 40 g/t and salinomycin at 60 g/t active compound).





ELSEVIER

Aquaculture Reports
Volume 42, 15 July 2025, 102745

Surfactin, bacillibactin and bacilysin are the main antibacterial substances of *Bacillus subtilis* JSHY-K3 that inhibited the growth of Vp_{AHPND} (the main pathogen of acute hepatopancreatic necrosis disease in shrimp)

Luoping Xiang [a b], Zijie Zhou [a b], Xiaotong Wang [b], Ge Jiang [a], Jie Cheng [a], Yadong Hu [c], Jin Qian [c], Xiaoman Sun [b], Hui Shen [a]

Article | Open access | Published: 25 September 2017

Use of antimicrobial peptides as a feed additive for juvenile goats

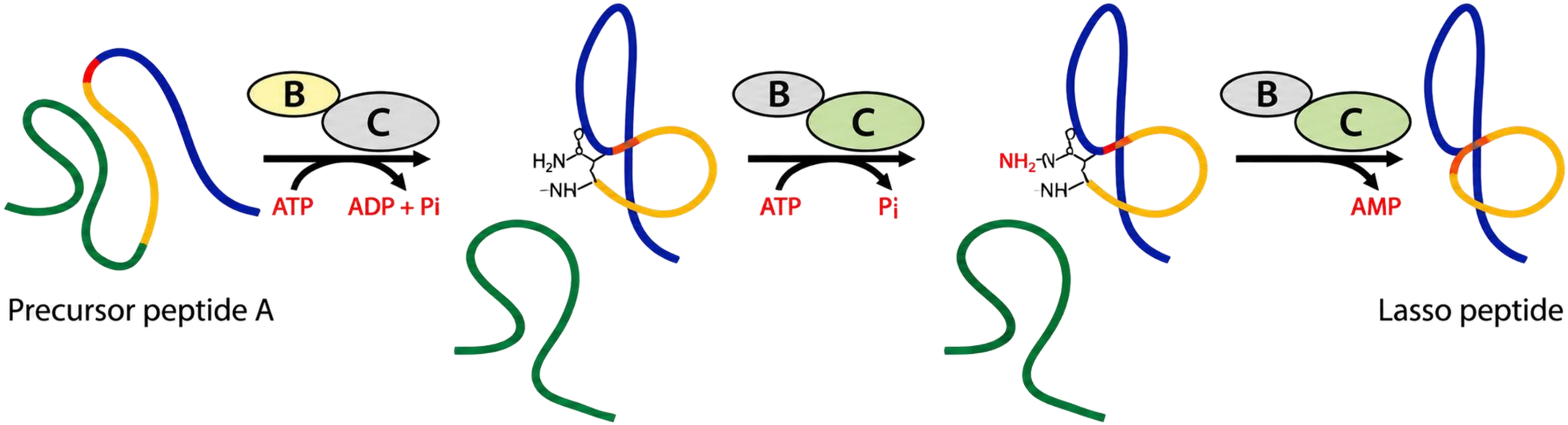
Qi Liu, Shuhua Yao, Yun Chen, Shuang Gao, Yanyi Yang, Junliang Deng ✉, Zhihua Ren, Liuhong Shen, Hengmin Cui, Yanchun Hu, Xiaoping Ma & Shumin Yu

Scientific Reports **7**, Article number: 12254 (2017) | Cite this article

In summary, this study showed that AMP supplementation maintained the rumen micro-ecological balance, but increased the relative abundance of *Fibrobacter*, *Anaerovibrio* and *Ophryoscolex*, and reduced the relative abundance of [*Paraprevotellaceae*]*CF231*, *Succinivibrio*, *Polyplastron* and *Isotricha*. The supplements also improved the rumen microbiota structure, altered ruminal fermentation, and increased the utilization efficiency of feed, thereby improving the potential growth performance. These results indicated that AMPs can be used as a feed additive in juvenile goats. Aranha *et al*[63]reported that

CITROCIN

a natural immunopeptide produced by *Citrobacter spp.* and *Saccharomyces spp.*



The Anti-Microbial Peptide Citrocin Controls *Pseudomonas aeruginosa* Biofilms by Breaking Down Extracellular Polysaccharide

Liyao Wang [1,2], Jiaqi Zheng [1,3], Wenchao Hou [1], Chaowen Zhang [1], Jie Zhang [1], Xuanbo Fan [1], Hongliang Zhang [1,4], Yuzhu Han [1,5,*]

Editor: Éric Biron

▸ Author information ▸ Article notes ▸ Copyright and License information

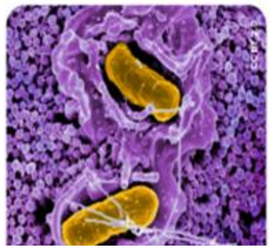
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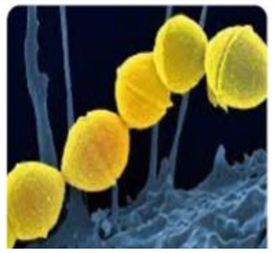
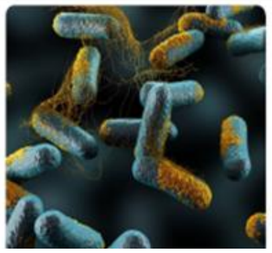
Abstract

Citrocin is an anti-microbial peptide that holds great potential in animal feed. This study evaluates the anti-microbial and anti-biofilm properties of Citrocin and explores the mechanism of action of Citrocin on the biofilm of *P. aeruginosa*. The results showed that Citrocin had a significant inhibitory effect on the growth of *P. aeruginosa* with a minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) of 0.3

Evaluation of Antimicrobial Efficiency

Products	Bacterial Pathogens			
	Escherichia coli	*Salmonella*	Staphylococcus aureus	*Clostridium perfringens*
Citrocin	✖✖	✖✖		
Surfactin			✖✖	✖✖



APPLICATIONS ACROSS ANIMAL PRODUCTION SYSTEM



POULTRY

Gut health
NE / enteric resilience



SWINE

Weaning stress
Enteropathogen pressure



RUMINANT

Digestive function
Production resilience

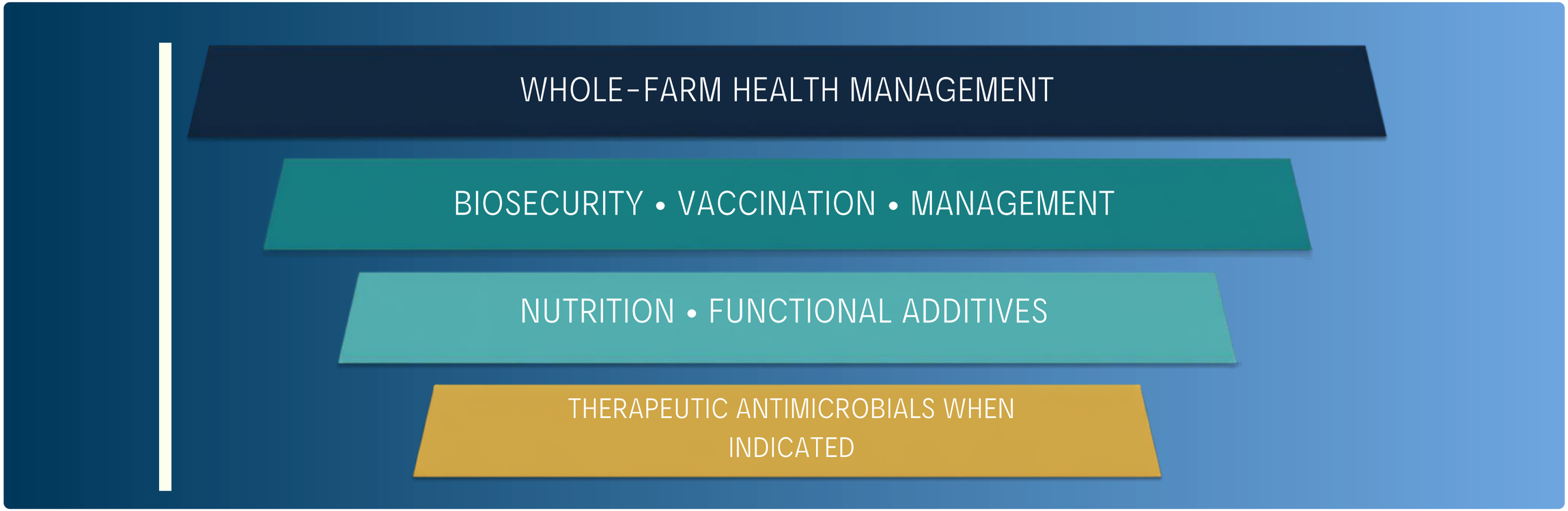


AQUACULTURE

Intestinal health
Immune robustness

Same concept ≠ same protocol. Species, age, diet, challenge, dose, and environment matter.

Immunopeptides and Responsible Antimicrobial Stewardship



Immunopeptides should complement—not replace—ANTIMICROBIALS.

TAKE HOME MESSAGE

1. HEALTHY GUT, HEALTHY ANIMALS
2. IMMUNITY IS ABOUT BALANCE
3. IMMUNOPEPTIDES BRIDGE HEALTH AND PERFORMANCE
4. SUSTAINABILITY STARTS WITH RESILIENCE

“Don’t just feed for growth—nourish the biology that makes growth possible.”

Thank you